Nikhil Goutham Budarayavalasa

Data Scientist

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SUMMARY

Data Scientist with 3+ years of experience in building machine learning models, developing data pipelines, and extracting insights from complex datasets. Expertise in supervised and unsupervised learning, deep learning, and natural language processing. Skilled in Python, SQL, and cloud-based data engineering solutions. Proven ability to design scalable AI models, optimize ETL workflows, and deploy data-driven solutions that enhance business decision-making.

SKILLS

Programming & Scripting: Python, R, SQL
Library: Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, Seaborn, Matplotlib, plotly, NLTK
Machine Learning & AI: Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Deep Learning, Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks, Natural Language Processing (NLP), Large Language Models (LLM), Time Series Analysis, Forecasting Models
Databases: PostgreSQL, MySQL, SQL Server, Google BigQuery, Amazon Redshift, Azure Synapse
ETL & Data Pipelines: Apache Airflow, Snowflake, Databricks
Cloud Platforms: AWS, Google Cloud Platform, Azure
Big Data Technologies: Apache Spark, Hadoop
Data Visualization & Analytics: Tableau, Power BI, A/B Testing, Hypothesis Testing, Bayesian Statistics
DevOps & Deployment: Docker, Kubernetes, CI/CD Pipelines, Git, Bitbucket, JIRA, Confluence
Soft Skills: Stakeholder Engagement, Team Collaboration, Strong Communication, Problem-Solving, Attention to Detail

EXPERIENCE

Data Scientist, Freddie Mac, USA — **September 2024 – Present**

- Built automated data warehouse from 12+ mortgage data sources, reducing processing time by 45% and enabling analytics for $800M loan portfolio.
- Developed and optimized ETL pipelines using SQL, and Apache Airflow to automate data processing and transformation.
- Led a project to fine-tune LLM for conversational AI applications, enhancing performance on benchmark datasets.
- Created ensemble ML models, CNN for mortgage risk prediction using Python, improving accuracy and supporting monthly lending decisions.
- Built initial RAG prototype integrating OpenAI API with internal documents for Q&A use cases.
- Built NLP document classifier for document processing, reducing manual review time and saving $25K annually in labour costs.
- Designed real-time dashboards for mortgage operations, improving decision speed and preventing potential losses monthly.

Data Scientist, Sage Softtech, INDIA — **January 2021 – December 2022**

- Created Power BI dashboards tracking 15+ business KPIs, providing visibility into $3M quarterly revenue and reducing reporting time.
- Optimized supply chain processes with ML models, increasing efficiency by 25% and reducing costs by $45K annually.
- Built data pipelines processing 300K+ daily transactions, improving query performance by 40% and enabling $2M revenue tracking capabilities.
- Deployed AWS-based NLP solution using spaCy and NLTK for automated text classification of customer feedback, processing 85% of documents automatically and saving 15 hours weekly
- Built machine learning models, including Logistic Regression, Decision Trees, XGBoost, Random Forest, and CNN to optimize business processes.
- Analyzed large datasets using Apache Spark, identifying cost optimization opportunities through statistical analysis and reporting.

Data Analyst, BHEL, Visakhapatnam, INDIA — **April 2019 – October 2019**

- Developed predictive maintenance models using Random Forest and ARIMA, reducing unplanned equipment downtime.
- Analyzed large-scale SCADA system data using SQL and Tableau to enable real-time plant performance monitoring.
- Created interactive dashboards to visualize operational insights, improving efficiency and decision-making.
- Tracked project progress and ensured timely delivery of insights using JIRA for project management.

EDUCATION

Masters in Data Science, New Jersey Institute of Technology, Newark, New Jersey — **January 2023 – December 2024**

PROJECT

Capstone Project – NJIT – Verizon, USA

- Enhanced operational monitoring and proactive decision-making by developing and deploying XGBoost models to detect patterns in fault logs. Improved data accessibility by processing and transforming large-scale unstructured JSON logs (50GB) in high-performance computing (HPC) environments using Python and advanced JSON flattening techniques.
- Aligned stakeholders on data-driven strategies by designing and maintaining Tableau dashboards for visualizing key insights.
- Optimized model selection for fault detection and enhanced HPC resource allocation by applying A/B testing,
- identifying the most accurate and computationally efficient model while determining the optimal configuration for processing large scale data.
- Improved Snowflake data models by collaborating with engineering teams and using Git for version control, ensuring seamless data integration and structured data organization.